

Hierarchical Semantic Memory Architecture (HSMA)

A Proposed Memory and Systems Architecture for Long-Horizon Agentic AI

Executive Summary

Large Language Models are rapidly evolving from single-request inference engines into persistent, long-horizon agentic systems capable of reasoning, planning, coding, tool use, and multi-agent collaboration. As these systems evolve, memory—not compute—is increasingly becoming the dominant systems challenge. Current approaches primarily optimize memory through: • KV cache compression • Retrieval-Augmented Generation (RAG) • Agent memory summarization • Knowledge graph retrieval This proposal explores a different architectural direction: **Treat memory as a hierarchy of understanding rather than a hierarchy of tokens.** The proposed Hierarchical Semantic Memory Architecture (HSMA) continuously transforms information through progressively more abstract representations.

Architecture Overview

Raw KV Cache → Compressed KV → Semantic Summaries → Concept Nodes → Knowledge Graph Rather than indefinitely storing all historical information in token form, the system continuously converts: **Tokens** → **Facts** → **Concepts** → **Understanding** while preserving the ability to selectively reconstruct detailed context when needed.

Problem Statement

Today's agentic systems increasingly exhibit: • Long conversations • Multi-step reasoning • Tool interactions • Persistent memory • Multi-agent collaboration These workloads create a fundamental scaling challenge: More interaction → More tokens → More KV state → More memory pressure Current memory systems remain largely reactive. They answer: "How do we store more tokens?" rather than: "How do we store more understanding?"

Limitations of Current Approaches

KV Compression • Reduces memory footprint • Preserves model behavior • No semantic structure RAG • Scalable external memory • Retrieval based on similarity • Weak representation of relationships Agent Memory • Compact summaries • Potential detail loss GraphRAG • Structured retrieval • Better semantic organization • Limited neural-state integration

Proposed Architecture

Level 0: Raw KV Cache Level 1: Compressed KV Level 2: Semantic Summaries Level 3: Concept Nodes Level 4: Knowledge Tree / Graph The key design principle: Every level points downward. Higher levels do not replace lower levels. They provide progressively more efficient representations.

Architectural Walkthrough

Example: France is in Europe. Paris is the capital of France. Berlin is the capital of Germany. Level 0: France, Europe, Paris, Berlin, Germany Level 1: Compressed Neural Memory Level 2: European countries discussed Level 3: Europe, Countries, Capitals Level 4: Geography ↓ Europe ↓ France ↓

Paris At this stage the system stores relationships rather than raw sequences.

Comparison Against Existing Approaches

KV Compression — Neural State — Memory reduction — No semantic structure RAG — Text Chunks — Retrieval scale — Weak relational understanding Agent Memory — Summaries — Compact memory — Detail loss GraphRAG — Concepts — Structured retrieval — Limited neural integration HSMA — Understanding Hierarchy — Multi-level memory abstraction — Significant research required

Proposed Hardware Architecture

Semantic Memory Controller Knowledge Cache Graph Memory Accelerator Hierarchical KV Compression Engine Semantic SSD Multi-Tier Agent Memory Fabric Knowledge-Aware Interconnect Reconstruction Accelerator Attention-Guided Demotion Engine Knowledge Processing Unit (KPU)

Why Agentic Systems Change the Memory Problem

Traditional inference workloads are bounded. Agentic systems may: • Persist for months • Coordinate with other agents • Build evolving world models • Continuously accumulate information Current systems optimize token storage. Future systems may need to optimize understanding storage.

Technical Advantages

• Reduced memory growth • Better long-term recall • Improved agent persistence • Potential reductions in HBM requirements • Reduced memory bandwidth demand • Lower retrieval overhead

Technical Risks and Open Research Questions

• Can abstraction preserve enough information? • Can lost detail be reconstructed accurately? • How should concepts be generated and updated? • How does the hierarchy evolve over millions of interactions? • What semantic operations justify dedicated silicon?

Research Roadmap

Phase 1: Software-only prototype Phase 2: Introduce compressed KV and reconstruction experiments Phase 3: Develop memory scheduling policies Phase 4: Explore hardware acceleration opportunities

Success Metrics

• Memory efficiency • Retrieval efficiency • Agent performance • Scalability to substantially longer interactions

Why This Could Become a New Systems Layer for Agentic AI

Current AI systems largely separate: Neural State Memory Knowledge HSMA proposes treating them as parts of a single hierarchy: Raw Computation Layer ↓ Memory Layer ↓ Knowledge Layer ↓ Understanding Layer If successful, the architecture could play a role similar to virtual memory, cache hierarchies, and storage hierarchies for long-horizon agentic intelligence.

Executive One-Page Summary

Proposal: Develop a Hierarchical Semantic Memory Architecture that transforms information through multiple abstraction layers: Raw KV → Compressed KV → Summaries → Concepts → Knowledge Graph Motivation: Current systems scale memory with token history. Future agentic systems may require memory that scales with accumulated understanding. Key Differentiator: Most existing approaches optimize storage of information. HSMA seeks to optimize storage of understanding. Hardware Opportunity: • Semantic Memory Controller • Knowledge Cache • Graph Accelerator • Reconstruction Engine • Knowledge Processing Unit Strategic Importance: If long-horizon agents become a dominant workload, memory management may become as important as model inference itself.